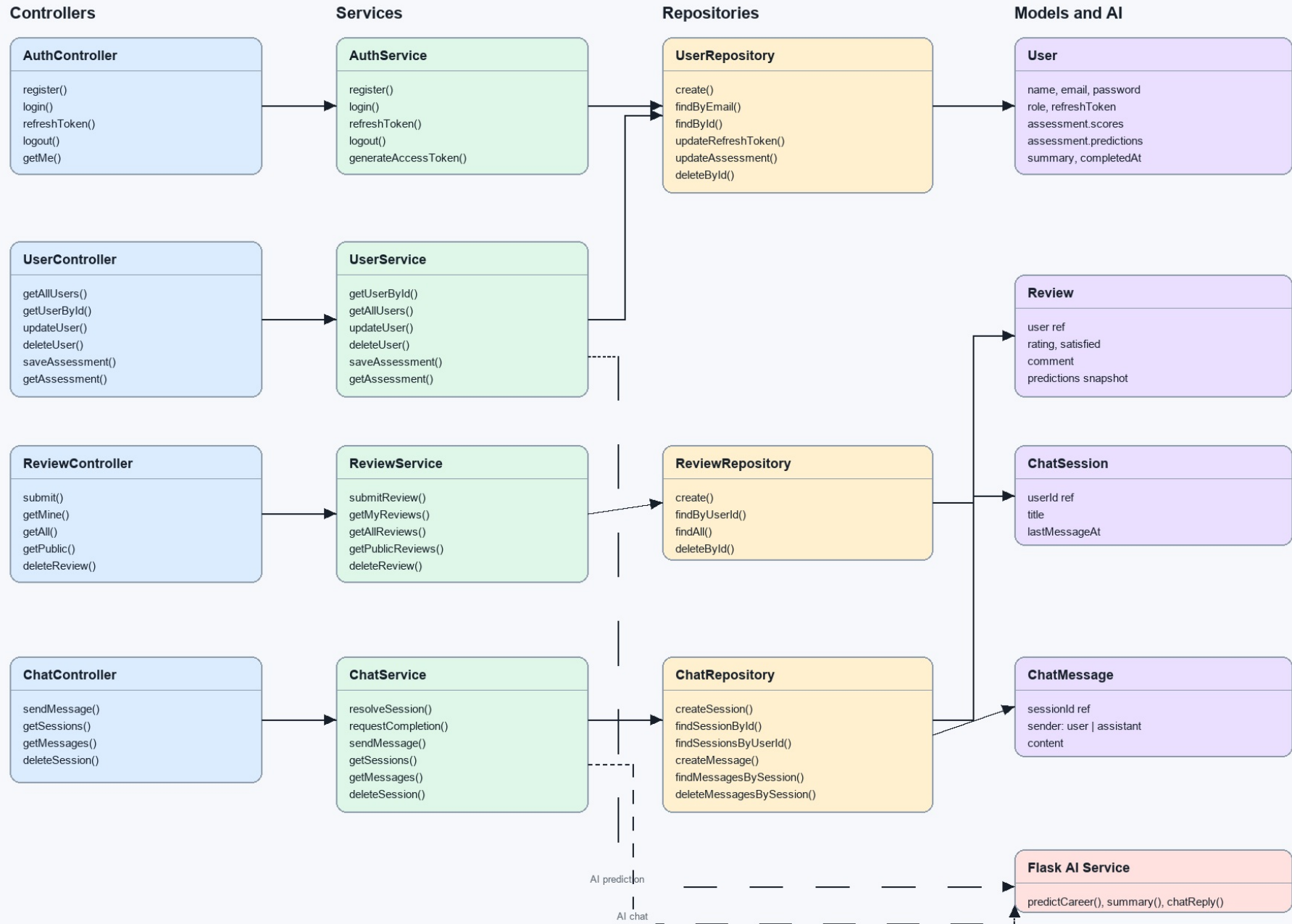


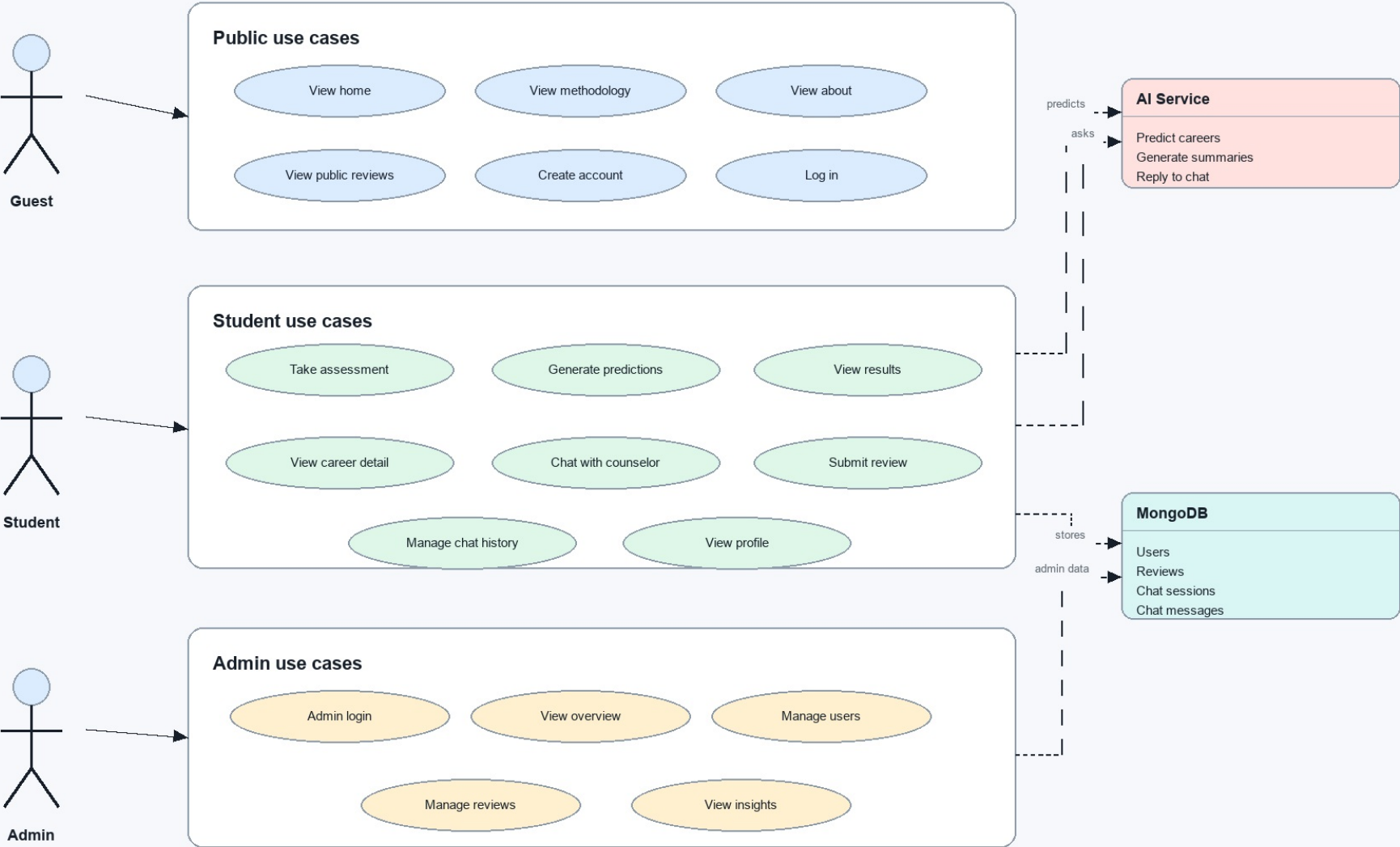
Backend Class Diagram

Controllers, services, repositories, data models, and AI service dependencies.



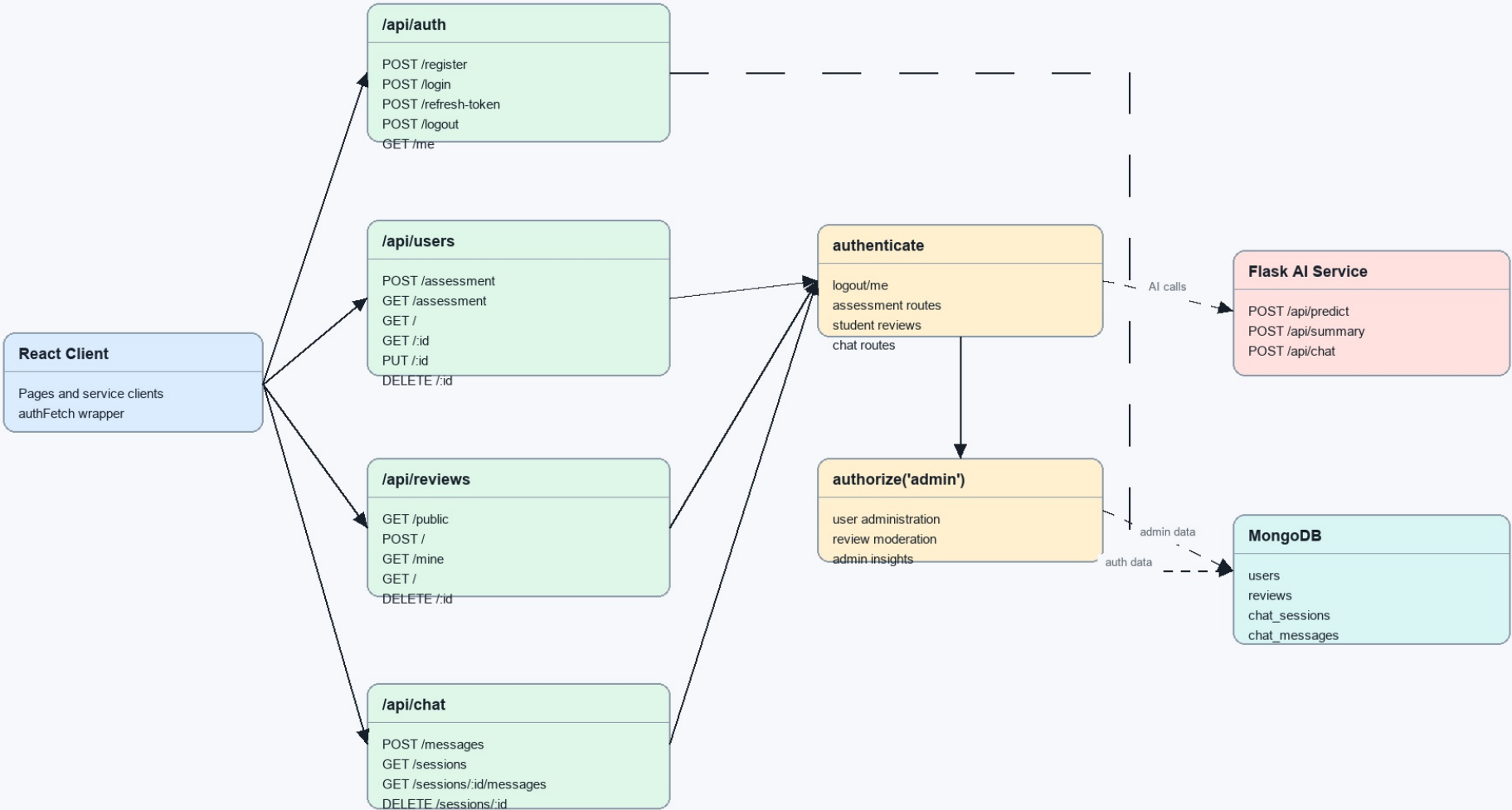
Use Case Diagram

Primary user goals for guests, registered students, admins, and external services.



API Use Case Diagram

React client route groups, authentication middleware, admin authorization, and AI service calls.



MongoDB Collection Schema Diagram

Collections, embedded documents, references, and cardinality in the Mongoose data model.

